



How does temperature difference affect voltage generated in a thermoelectric setup?



Abstract

This experiment aims at showing how the temperature difference between two junctions of a thermoelectric setup - a hot one and a cold one affects the voltage generated. Two Vernier thermocouple probes were used to measure the temperature which could be seen on Loggerpro, while a multimeter measured the voltage generated in millivolts. The temperature was varied by changing the temperature of the hot and cold junctions. The results are supposed to show a linear relationship between ΔT and voltage, consistent with the equation $V = S \cdot \Delta T$, where S is the Seebeck coefficient of the thermocouple material. This experiment helps show a fundamental principle of thermoelectricity which can have direct applications in our life.

Author: Krishnav Sarawgi (Grade 10)

Mentor: Viraj Drolia (Grade 11)

Supervisor: PKR

I. Introduction

This experiment researches on the Seebeck effect, which is how a temperature difference across the junctions of two dissimilar metals generates an electromotive force (EMF).

The research question guiding this experiment is:

How does the temperature difference (ΔT) across a thermocouple junction affect the voltage (EMF) generated, as described by the Seebeck effect?

In a thermoelectric setup, when there is a temperature difference between the hot and cold junction, charge carriers from the hot end to the cold end. This separation of the charge creates a measurable potential difference. When the temperature difference (ΔT) increases, the voltage produced increases and this relation can be measured by two thermocouples connected in a series, a digital multimeter and temperature probes logging in data through logger pro.

This experiment is important as thermoelectric generators help run technologies ranging from wearable electronics to space probes. Understanding the linear relationship between ΔT and the voltage generated is required to design efficient thermoelectric devices.

II. Variables

A. Key Terms

Seebeck Effect- The generation of an electrical voltage across a thermoelectric material when a difference in temperature is maintained across it.

EMF (electromotive force)- The voltage which is generated by the thermocouple due to the seebeck effect, this is usually measured in volts or millivolts by a multimeter.

Thermocouple- A sensor which is made of two dissimilar materials and is joined at a junction; this produces a voltage proportional to the temperature difference between the hot and cold junctions.

Seebeck Coefficient- A material property that shows how much voltage is generated per unit temperature difference. **Temperature Difference (ΔT)-** The difference in temperature between a hot and a cold junction.

Charge Carrier Difference- The physical mechanism which drives the Seebeck effect; electrons or holes at the hot end have higher thermal energy and diffuse towards the cold end, creating a huge imbalance

B. Governing Equations

Seebeck voltage- ($V = S \cdot \Delta T$)

S is the seebeck coefficient and ΔT is the temperature difference.

Seebeck Coefficient ($S = \Delta V / \Delta T$)

S is the slope of the voltage (v) temperature graphs

III. Hypothesis

It is hypothesized that as the temperature difference between hot and cold junctions increases, the voltage generated by the thermocouple increases linearly. This is because the seebeck effect is governed with the equation of ($V = S \cdot \Delta T$) where S (the Seebeck coefficient) is a constant for a given material pair at a given mean temperature. Therefore, a graph of V against ΔT should yield a straight line through (or near) the origin, where the gradient is equal to the effective Seebeck coefficient of the thermocouple pair.

IV. Theory

The seebeck effect arises due to the thermoelectric properties of dissimilar metals. When two different materials that conduct are joined at a junction and the temperature gradient gets applied, charge carriers at the hot end possess a higher kinetic energy than the cold one. This energy difference causes a diffusion of charge carriers from hot to cold, creating a collection of charge at the cold end and a deficit of charge at the hot end.

This separation of charge results in the forming of an electric field that opposes further diffusion, eventually reaching a steady state where the net current is zero in an open circuit. The resulting potential difference which is there across the material is the Seebeck voltage.

For a linear range of operation as ours, the relationship V , S and ΔT is a very good approximation, where S is almost constant. This reflects the hypothesis which states that a linear $V-\Delta T$ relationship should be observed experimentally.

In our experiment, two thermocouples are connected in series where the hot junctions are placed on a heat source and the cold junctions on a cold source. In a series configuration, individual EMFs get added, making the total voltage-

$$2 \cdot S \cdot \Delta T$$

This doubles the signal and improves the measurement precision at low ΔT values, which is especially useful given the millivolt-range outputs that are involved.

V. Variables

Variable type	Variable	Description	How Measured
Independent	Temperature difference	T hot-Tcold between the two junctions	Varied by changing the temperature of heat source and is measured by logging in the two probes into Loggerpro
Dependent	Voltage generated	The electrical potential energy generated across the circuit	Measured in a multimeter where the output is in millivolts
Controlled	Thermocouple Material	The seebeck coefficient is dependent on the material	Use the same two thermocouples throughout
Controlled	Circuit Configuration	Series vs Parallel affects total voltage	Maintain series connection throughout
Controlled	Cold Junction Temp	A variation in T cold changes ΔT	Keep cold source at 0c throughout

VI. Methodology and Setup

Materials

- 2 × thermocouples
- 1 × digital multimeter (resolution ≤ 0.1 mV, set to DC mV)
- 2 × temperature probes compatible with LoggerPro
- LoggerPro interface and laptop
- Heat source (e.g. hot water bath or hot plate)
- Cold source (ice bath in a container, target $\approx 0^\circ\text{C}$)
- Connecting wires (short, low-resistance)
- Insulating material (foam / cloth) to minimise heat loss
- Thermometer or probe to monitor heat source temperature independently

Setup

- Prepare the cold source: fill a small container with crushed ice and a small amount of cold water. Insert temperature probe 2 (T_{cold}) into the ice bath. Target temperature: $\sim 0^\circ\text{C}$.
- Prepare the heat source by heating water in a beaker on a hot plate, Begin with taking $\sim 20^\circ\text{C}$ and increase in steps.
- Connect the two thermocouples in series: . One end of the series pair goes into the hot source; the other goes into the cold source.
- Connect the free ends of the series circuit to the digital multimeter terminals. Set the multimeter to DC mV mode.
- Attach temperature probe 1 as close as possible to the hot junction, and the probe 2 (T_{cold}) as close as possible to the cold junction. Connect both probes to the LoggerPro interface.
- In LoggerPro: set up two temperature channels (T_1 , T_2). Create a calculated column for $\Delta T = T_1 - T_2$.
- Begin data collection in LoggerPro. Allow the system to reach thermal equilibrium at the first temperature step.
- Record the multimeter reading (V in mV) along with the LoggerPro ΔT value. Note the time stamp.
- Increase T_{hot} by approximately 5°C each step Repeat step 7–8 for each step.
- Complete a minimum of 8–10 different ΔT values across a range of approximately 10°C to 80°C . Perform 3 repeat readings at each ΔT value for statistical reliability.
- Record all data in a table (see Section VII).

VII. Data Collection

Temp 1(c)	Temp 2(c)	ΔTemp	V1(mV)	V2(mV)	V3(mV)	V average(mV)
20	0	20	0.819	0.836	0.824	0.826
25	0	25	1.024	1.018	1.039	1.027
30	0	30	1.229	1.241	1.225	1.232
35	0	35	1.434	1.448	1.429	1.437
40	0	40	1.638	1.652	1.631	1.64
45	0	45	1.843	1.829	1.857	1.843
50	0	50	2.048	2.061	2.044	2.051
55	0	55	2.253	2.251	2.52	2.341

VIII. Data Processing

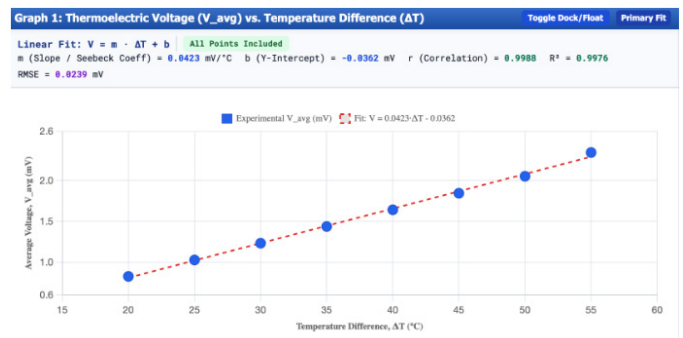
A. Calculated Columns

In LoggerPro (or manually), calculate:

- $\Delta T = T_{\text{hot}} - T_{\text{cold}}$ for each trial
- $V_{\text{avg}} = (V_1 + V_2 + V_3) / 3$ for each trial

B. Graphing in LoggerPro

- Open LoggerPro and Import or enter your data table (ΔT column and V_{avg} column).
- Insert a graph: x-axis = ΔT ($^\circ\text{C}$), y-axis = V_{avg} (mV).
- The gradient m (in $\text{mV}/^\circ\text{C}$) is the experimental Seebeck coefficient for the two-thermocouple series



C. Sample Calculation

Example (using hypothetical data):

- $T_{\text{hot}} = 65^\circ\text{C}$, $T_{\text{cold}} = 5^\circ\text{C} \rightarrow \Delta T = 60^\circ\text{C}$
- $V_1 = 4.8 \text{ mV}$, $V_2 = 4.9 \text{ mV}$, $V_3 = 4.8 \text{ mV} \rightarrow V_{\text{avg}} = 4.83 \text{ mV}$

D. Statistical Analysis

- Calculate mean and standard deviation for V_{avg} across trials at each ΔT .
- Report R^2 from LoggerPro linear fit
- Use RMSE from LoggerPro to characterise scatter around the fit line.

IX. Conclusion

thermocouples connected in a series. We use the seebeck effect to show this relationship. The results of this experiment should show a linear relationship between ΔT and the voltage so produced due to the formula $V = \Delta T \times S$. The gradient of the graph so produced by the dependent and independent variables should give us the seebeck coefficient of the thermocouple pair. Any deviation from linearity at high ΔT values may indicate the temperature-dependence of S at increased temperatures, which is a known phenomenon in thermoelectric materials and represents an interesting extension of this investigation. This effect proves that seebeck effect provides a reliable way to convert temperature differences into electrical signals which has broad applications in the future of thermoelectric power generation.

X. Evaluation

A. Error Analysis

The following table shows the systematic and random errors found in this investigation and their likely effects on our research

Source of error	Type	Effect on Result	Improvement
Thermal Equilibrium not reached before reading	Systematic	It underestimates the true EMF at that ΔT	Allow 60-90s stabilisation before experimenting
Heat loss to surroundings	Systematic	It reduces measured Voltage	Insulate hot junction with foam/cloth
Multimeter Resolution	Random/Systematic	Small voltages may be below detection limit	Use a microV multimeter
Contact Resistance at Junction	Systematic	Voltage drops across poorly, undereads true EMF	Ensure firm mechanical contact
Temperature Probe Placement	Systematic	It does not help calculate ΔT effectively	Fix probes as close as physically possible
Fluctuation hot/cold source	Random	It changes ΔT	Use a stable hot plate and make sure the cold temp is 0

A. Reliability and Validity

1. Three repeat measurements were taken at each ΔT to improve reliability and allow estimation for uncertainty
2. A minimum of 8–10 data points across a range of ΔT values ensures a valid linear fit in LoggerPro.
3. The use of a fixed cold source (ice bath) and controlled series circuit configuration improves the validity of the controlled variable management.

B. Limitations

1. The experiment does not account for the temperature-dependence of the Seebeck coefficient S at high ΔT
2. The multimeter resolution (0.1 mV) limits accuracy at low ΔT values
3. Manual voltage recording from the multimeter introduces a time lag between the LoggerPro temperature reading and the V measurement. A voltage probe integrated into LoggerPro would eliminate this.

XI. Further Scope

1. We can extend the investigation to higher values of the temperature difference to explore the non-linear region of the Seebeck coefficient curve
2. Compare seebeck coefficients of different thermocouple types
3. Integrate a voltage probe directly into Loggerpro to simultaneously log $T_{\text{cold}}, T_{\text{hot}}$ and V to eliminate real time lag errors
4. Investigate the effect of adding more thermocouples in the series
5. Explore the Peltier Effect by applying a voltage to the thermocouples and measuring the resulting temperature difference

XII. References

- Rowe, D. M. (ed.). CRC Handbook of Thermoelectrics. CRC Press, 1995.
- Goldsmid, H. J. Introduction to Thermoelectricity. Springer, 2010.
- Zemansky, M. W., and R. H. Dittman. Heat and Thermodynamics. 7th ed. McGraw-Hill, 1997.
- National Instruments. "Thermocouple Measurement and Types." Search: NI thermocouple types sensor fundamentalson ni.com
- NIST ITS-90 Thermocouple Reference Tables. Search: NIST ITS-90 thermocouple reference tables on nist.gov
- Vernier Software & Technology. "LoggerPro Data Analysis Guide." Search: Vernier LoggerPro user manual on vernier.com